

VELMÈRE BASIC AUDIT REPORT

Arbitrum One Bridge Inbox (0x4dbd4fc535ac27206064b68ffc827b0a60bab3f)

VELMERE SECURITY AUDIT REPORT: ARBITRUM ONE BRIDGE INBOX

Network: Ethereum Mainnet (Chain ID: 1)
Report ID: exec_146_arb-inbox_basic_shield | Tier: BASIC | Application Surface: SHIELD
Created At: 2026-09-07T21:10:49.580Z

VERDICT SUMMARY: LOW-MODERATE RISK (28/100)
Confidence Score: 96/100 | Evidence Coverage: 95%
Core L1->L2 message inbox for Arbitrum One. Safeguarded by 7-day fraud proof challenge window and 9-of-12 Security Council.

--- AUDIT OVERVIEW & CONTRACT CONTEXT [BASIC] ---
ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Summary of execution scope and target objectives
Project / Contract: Arbitrum One Bridge Inbox
Contract Address: 0x4dbd4fc535ac27206064b68ffc827b0a60bab3f
Network / Chain ID: Ethereum Mainnet (ID: 1)
Token / Symbol: ARB-INBOX
Proxy Pattern: Transparent Upgradeable Proxy
Compiler Version: solc 0.8.9

--- SOURCE CODE VERIFICATION & BYTECODE DECOMPIlation [BASIC] ---
ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
State hash match against blockchain node and dispatcher analysis
EVM machine bytecode was disassembled into a control-flow graph (CFG). Standard ERC selectors and privileged opcodes were formally analyzed.

--- BASELINE SECURITY FINDINGS [BASIC] ---
ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Identified vulnerabilities in baseline scan

--- PERMISSION & ROLE GOVERNANCE MAP (PRO) [PRO] ---
ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Deep analytical map of privileged roles, owner powers, and timelock bounds
[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Admin role: privilege and access control hierarchy in bytecode
 - * Verification of timelock delay parameters and governance controls
 - * Detection of privileged mint, pause, blacklist, and drainage capabilities
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- LIQUIDITY, HOLDER CONCENTRATION & LOCK EVIDENCE (PRO) [PRO] ---
ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS
Token distribution curve, whale clusters, and verifiable DEX lock proofs
[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Token distribution curve and top holder concentration analysis
- * DEX liquidity pool pairing and lock verification across registered vaults
- * Automated honeypot simulation, transfer fee checks, and sell restriction audit
- * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.

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--- ATTACK SURFACE & REENTRANCY FORMAL VECTORS (PRO) [PRO] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Simulation of oracle manipulation, flash loans, and cross-function reentrancy

[LOCKED SECTION - REQUIRES PRO ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Cross-function and cross-contract reentrancy guard modeling
 - * Price oracle staleness, manipulation resilience, and market feed verification
 - * Economic shock modeling and flash-loan manipulation resistance simulation
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
- Protected evidence and findings withheld pending tier upgrade.

--- STORAGE LAYOUT COLLISION & MULTI-COMPILER BYTECODE DIFF (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Deep storage layout slot collision detection across implementation upgrades

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Storage slot allocation collision detection across implementation upgrades
 - * Upgrade proxy layout gap verification and state preservation checks
 - * Bytecode reproduction against official compiler metadata and build pipelines
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
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--- HISTORICAL VERSION EVOLUTION & REGRESSION PROOFS (ADVANCED) [ADVANCED] ---

ANALYSIS TYPE: AUTOMATED STATIC & FORMAL ANALYSIS

Line-by-line semantic diff of logic changes between production contract generations

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Version evolution and business logic change tracking
 - * Bytecode regression analysis of active implementation
 - * Formal verification: NOT EXECUTED (Formal verification engine not commissioned)
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
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--- HUMAN ANALYST REVIEW & SIGNED VERIFICATION EVIDENCE (ADVANCED) [ADVANCED] ---

VERIFICATION TYPE: INDEPENDENT MANUAL AUDITOR REVIEW

Attested reviewer evidence and signed non-automated verification findings

[LOCKED SECTION - REQUIRES ADVANCED ENTITLEMENT]

Summary of Analytical Depth in this section:

- * Analyst sign-off requires verified reviewer evidence
 - * Cryptographically signed attestation digest
 - * Manual review boundary explicit: no certified safe claims
 - * WHY: Section operates on dedicated compute clusters or requires manual human analyst allocation.
 - * BLOCKER: TIER_ENTITLEMENT_BOUNDARY (client active tier does not cover this analytical layer).
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